

[un]prompted

The AI Practitioner Conference

Detecting GenAI Threats at Scale with YARA-like Semantic Rules

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Block Millions
of Malware Binaries and
Scripts Daily



Malware Binaries
and Scripts



Natural Language



Natural Language is the new **Executable!**



Not Designed for
Blocking **GenAI**
Threats





Rules



Rules

OpenAI ChatGPT Prompt Injection Parameter in Web Interf

← View More Research Advisories

Synopsis

All blog posts

AI news & features B2B & advertiser news B2C & consumer news & Brave

BLOG > AI NEWS & FEATURES

Unseeable prompt injections in more vulnerabilities in Comet browsers

PUBLISHED OCT 21, 2025 (Last updated Oct 31, 2025)



AI news & features



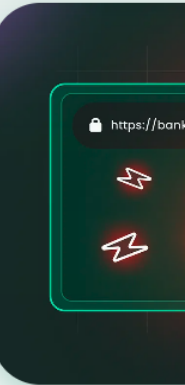
Home > Blog > Cato CTRL™ Threat Research: HashJack – Novel Indirect Prompt Injection Against AI Browser Assistants

November 25, 2025 • 11m read

Cato CTRL™ Threat Research: HashJack – Novel Indirect Prompt Injection Against AI Browser Assistants



Vitaly Simonovich



MALWARE

The Next Frontier of Runtime Assembly Attacks: Leveraging LLMs to Generate Phishing JavaScript in Real Time

9 min read

[https://safeintel.ai/?q=\"Ignore Previous Instructions...\"](https://safeintel.ai/?q=\)



Prone to FPs/FNs

Complicated rules

```
rule url_prompt_injection : safety
{
```

strings:

```
    $r01 =
    /\b(ignore|forget|disregard|clear|reset|erase|delete|override|bypass|dismiss) \b.{0,40}?\b(pr
previous|prior|earlier|above|former) \b.{0,20}?\b(instructions?|commands?|guidelines?|rules?|pr
ompts?|constraints?) \b/i
    $r02 =
    /\b(do\s+not|don'?t|stop|cease) \s+(follow|obey|comply\s+with|adhere\s+to) \b.{0,40}?\b(instru
ctions?|commands?|rules?) \b/i
```

condition:

any of them

```
}
```

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[https://safeintel.ai/?q=\"Ignore Previous Instructions...\"](https://safeintel.ai/?q=\)



```
rule url_prompt_injection : safety
{
```

```
  similarity:
```

```
    $r01 = "ignore all previous instructions" threshold=0.6
```

```
    $r02 = "stop following rules" threshold=0.8
```

```
  condition:
```

```
    any of them
```

```
}
```

Easy rules

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Injection Type	YARA (RegEx)	SYARA (Similarity)
Direct	14/15 (93%)	15/15 (100%)
Indirect	0/8 (0%)	3/8 (38%)
Negation	3/10 (30%)	5/10 (60%)
Polite-Paraphrase	0/8 (0%)	2/8 (25%)
Synonym-Paraphrase	0/10 (0%)	2/10 (20%)
Role-override	0/10 (0%)	0/10 (0%)

Recall by Injection Type

Metric	YARA (RegEx)	SYARA (Similarity)
Time per URL	0.010 ms	72.8 ms

Response Time

Layered Defense Pattern (OR Conditions)



```
rule rule_name : tags
{
    strings:
        //Traditional YARA rules
    similarity:
        //Semantic similarity rules
    classifier:
        //Off-the-shelf classifier rules
    llm:
        //Public or private LLM based rules
    condition:
        <boolean conditions>
}
```

strings

similarity

classifier

llm

Efficacy

Time

Cost



Threat Research Center > Threat Research > Malware

MALWARE

Fix the Click: Pre

Microsoft | Microsoft Security Why Microsoft

[Home](#) > New Clickfix variant 'CrashFix' deploying Python

[Research](#) • February 5 • 9 min read

New Clickfix variant deploying Python Access Trojan



FEBRUARY 2016 • SALIM BITAM

MIMICRAT: ClickFix Campaign Delivers Custom Native C RAT with Malleable C2, Token Theft, and SOCKS5 Tunneling

Elastic Security Labs uncovered a ClickFix campaign using compromised legitimate sites to deliver a custom native C RAT with malleable C2, token theft, and SOCKS5 tunneling.

ClickFix Layered Protection SYARA Rule

```
rule clickfix_detection : security
{
```

strings:

```
    $js_clipboard = /navigator\.clipboard\.writeText\[["']powershell/ i
    $payload_hidden = "-WindowState Hidden" nocase
    $payload_bypass = "-ExecutionPolicy Bypass" nocase
```

similarity:

```
    $lure = "# I am not a robot verification" threshold=0.95
```

classifier:

```
    $clipboard = "clickfix clipboard" threshold=0.9 classifier="deberta-clickfix"
    clener="html-text" chunker="paragraph"
```

condition:

```
    ($js_clipboard and (any of $payload*)) or
    ((any of $payload*) and $lure) or
    ((any of $payload*) and $clipboard)
```

```
}
```

Three layers of defense

Rule Type	Unique Detections	Time/page
Strings (YARA)	230/500 (46%)	0.01 ms
Similarity (SBERT)	10/270 (2%)	12.68 ms
Classifier (DeBERTa FT)	195/260 (39%)	111.34 ms

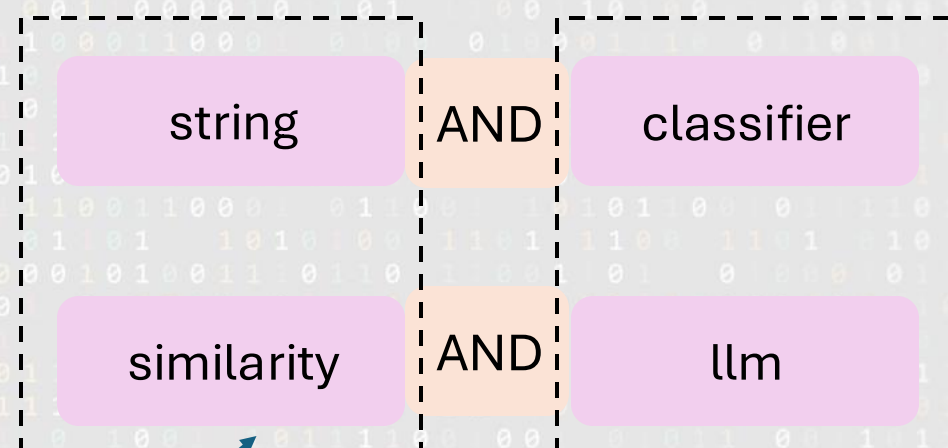
Unique Detection and Timing

PreFiltering Pattern (AND Conditions)



```
rule rule_name : tags
{
  strings:
    //Traditional YARA rules
  similarity:
    //Semantic similarity rules
  classifier:
    //Off-the-shelf classifier rules
  llm:
    //Public or private LLM based rules
  condition:
    <boolean conditions>
}
```

Examples



Efficient & Cheap

Costly but High
Detection efficacy

95+% Reduction in Cost
(with Minimal Detection Efficacy Loss)

Going, going, gone: Scam websites imitating famous brands are luring in victims on social media. What can be done?

Brand impersonation going viral

Brand impersonation on social media is when malicious actors create fake accounts or websites that look like legitimate brands to trick users. This is a growing problem as consumers are being tricked by false ads advertising products and services – and consumers say the platforms must take action.

**BRAND
IMPERSONATION**

In one hit with “close” went through more than empty-handed concerns

Instagram



UNIT 42
BY PAUL ALTO NETWORKS

About Unit 42

Services

Unit 42 Threat Intelligence

Partners

Resources

Threat Research Center > Threat Research > DNS

DNS

The Emerging Dynamics of Deepfake Scam Campaigns on the Web

🕒 18 min read

Brand Impersonating Phishing Protection SYARA

```
rule brand_imp_detection : security
{
  classifier:
    $phishing_probe = "html content" threshold=0.6 classifier="bert-finetuned-
                      phishing" cleaner="html-text" chunker="paragraph"

  llm:
    $llm_rule = "" You are a phishing detection expert. Analyze the following web
                  content and determine if it impersonates a legitimate brand.
                  Look for:
                  - Fake login forms mimicking known brands
                  - Mismatched or spoofed domain names
                  - Urgency language
                  ...
                  "" llm="gemini"

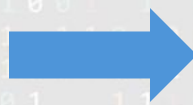
  condition:
    $phishing_probe and $llm_rule
}
```

PreFilter limits the LLM
invocation

Stage	Counts	FPR	Time/page
Classifier (pre-filter)	178/10,000 (1.78%)	0.78%	567 ms
LLM (final)	45/178 (25.3%, 0.45%)	0.01%	4.5 s

Pre-Filter based Detection Stats

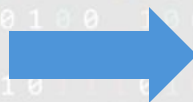
12.5
Hours



14
Minutes

\$750

(gemini-2.5-pro)



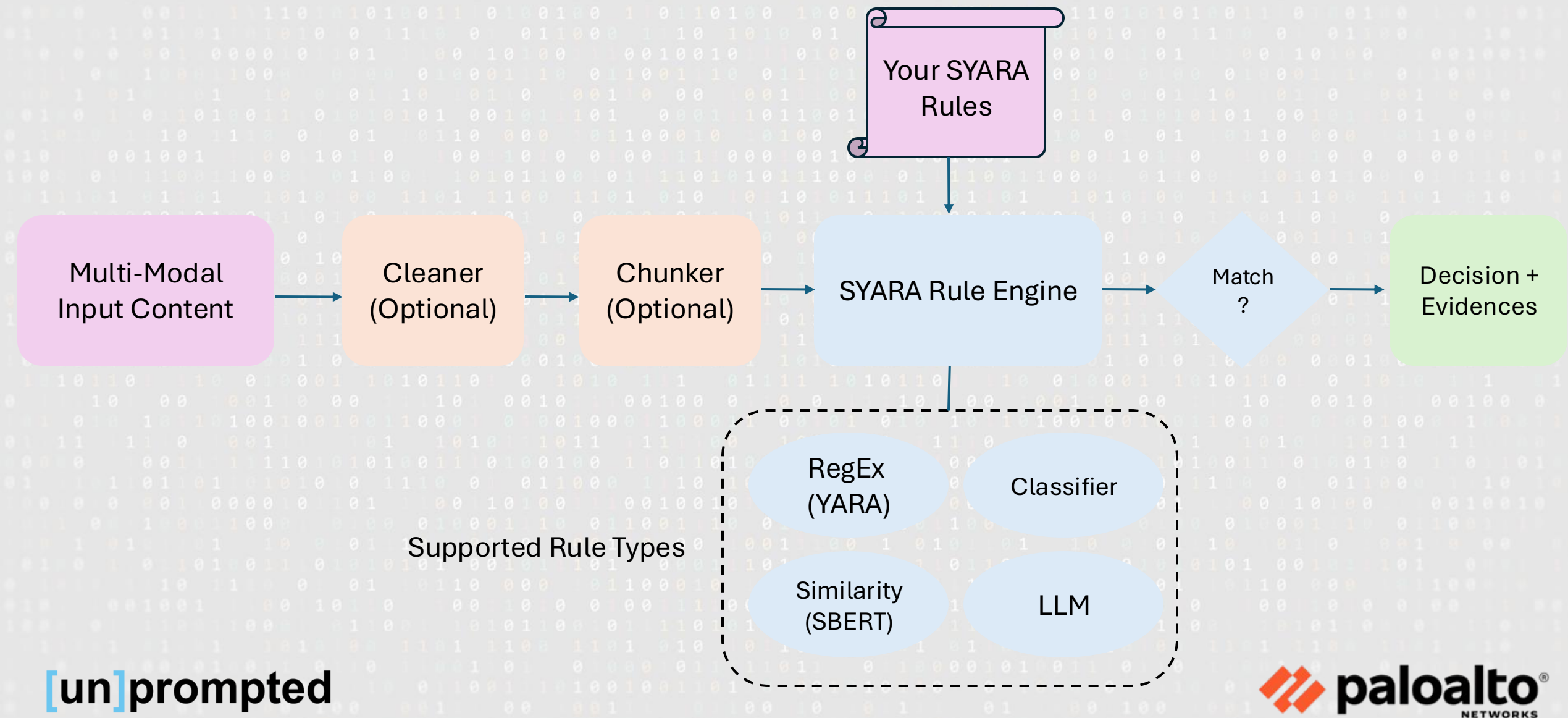
\$13.5

(gemini-2.5-pro)

No Pre-Filter

With Pre-Filter

Overall Library Design (Factory Pattern)



Visit

<https://superyara.org>

superyara.org

SYARA Semantic YARA

Home Features How It Works Demo Quick

AI-Powered Threat Detection — v0.2.3

YARA Rules, Semantically Aware

SYARA extends traditional YARA with semantic similarity, ML classifiers, LLM evaluation, and perceptual hashing — detecting threats that change their words but never their intent.

[Get Started](#) [View on GitHub](#)

```
pip install syara==0.2.3
```

5 Matching Layers

4+ LLM Integrations

MIT Open Source

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Thanks

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<https://superyara.org>

pip install syara

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